

Statistics 652: Advanced Theory of Statistics

Fall 2026 Section: 01 Index Number: 20837

Room: Hill 552 (Busch Campus)
Time: Tuesdays and Thursdays
 3:50 - 5:10 PM (5th period)

Instructor: Dr. David Tyler
Office: 567 Hill Center (Busch Campus)
E-mail: dtyler@stat.rutgers.edu
Web-site: dtyler-rutgers.github.io (under construction)

Office Hours: Tuesdays 5:15 - 6:00 PM
 Other times by appointment.

Grading Policy: Homework Problems = 50%.
 Quizzes may be given in lieu of homework problems.
 Class Participation, Class Projects and Presentations = 50%
 Note: Grading policy is subject to change.

Syllabus: A *tenative* list of topics to be covered are listed below. The syllabus will be adapted as the course proceeds.

- Convexity and the theory of exponential families.
- Information and univariate large sample theory.
- Topics on multivariate and multiparameter statistics.
 - Multivariate distributions and probability measures.
 - Visualizing multivariate data.
 - Principal components, canonical components and the spectral value decomposition.
 - Finite sample and large sample multivariate distribution theory.
 - Normal graphical models and penalization methods.
- Basic topics in robust statistical methods.
 - Robust functionals and the influence function.
- Basic topics in functional data analysis.
 - “PCA” on Hilbert spaces.

Course text: References to outside reading will be given as needed.

Homework: You are expected to keep a neat folder of all your homework assignments, which *may* be collected at the end of the semester. Some homework assignments will be collected periodically and sometimes graded. Not all homework assignments will be collected, and not all homework assignments which are collected will be graded.